Age confirmation of participants

Read me when using external libraries

Memory Limit(MB): 512, Execution Timeout(ms): 2000

You will participate in a meetup with your fellow hobbyists. You are in charge of organizing the after-party for this meetup. In your country, the age considered to be of legal adulthood is (x), and anyone under the age of (x) is considered a minor and is not allowed to drink any beverages containing alcohol.

For the after-party, the group will order an all-you-can-drink course. If there are any underage participants at the after-party, everyone will order the all-you-can-drink course that includes only soft drinks. On the other hand, if there are no underage participants, everyone will order the all-you-can-drink course that includes alcoholic beverages.

Since it would be a hassle to go through the age verification process on-site, you decided to check the list of ages of the participants in advance to figure out the all-you-can-drink course you should choose.

Given the number of participants n, the age at which a participant will be considered to be of legal age x, and a list of participants' ages $y_1, \ldots, y_n$, create a program to output whether the all-you-can-drink course will include alcohol or be limited to only soft drinks.

Implementation Details

CLI

Implement a command line application that receives standard input and writes the answer to its standard output. For details, see the "Command line application template" section at the bottom of the page.

Input Rules

The program is executed as follows:

```
./myApp < input.in
```

Implementation Details

CLI

Implement a command line application that receives standard input and writes the answer to its standard output. For details, see the "Command line application template" section at the bottom of the page.

Input Rules

The program is executed as follows:

```
./myApp < input.in
```

The standard input format is as follows:

$$n \ x$$
$$y_1 \ y_2 \ \cdots \ y_n$$

In the first line, a space-separated list of the number of participants n, and the age at which they are considered to be of legal age x, are given.
In the second line, a space-separated list of the ages of the participants is given, $y_i (1 \leq i \leq n)$.

Restrictions:

- $1 \leq n \leq 100$
- $10 \leq x \leq 21$
- $10 \leq y_i \leq 120 (1 \leq i \leq n)$
- All inputs are integers.

Output Rules

r CLI program should write the answer to its standard output.

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Age confirmatio

Age confirmation of participants

Output Rules

Your CLI program should write the answer to its standard output.
The format of the standard output is as below:

- If there are any values in the list of ages that are less than the age considered to be of legal adulthood, output "Soft Drink".
- If there are no values under the age considered to be of legal adulthood in the list of ages, output "Alcohol".

I/O Examples

Example 1

Standard input (`00_sample_01.in`)

```
5 20
25 23 28 19 20
```

Standard output

```
Soft Drink
```

Output "Soft Drink" because the list of ages includes participants whose age is under 20 years old.

Example 2

Standard input (`00_sample_02.in`)

```
3 18
120 18 65
```

Standard output

Standard output

```
Alcohol
```

Some expected inputs and outputs are defined in test/basic_testcases.json. Please refer to them when implementing.

Command line application template for C++

Implement CLI application by editing main.cpp.
You may add new files to keep your code clean, if it is allowed in your challenge.

How to get stdin lines

To get data from stdin, you can use an object `cin` or a function `getline` in the `std` name space.
The template code has example to read all stdin lines.

You can modify that code as you like.

How to output result

You can use `cout` , `printf` , etc.

```
cout << argv[0] << endl
```

Copy

How to compile

To compile, we are using clang g++ command.

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How to compile

To compile, we are using clang g++ command.

Install External Libraries

Libraries not included by default can be made available by installing them using `apk add` within the makefile.

Never execute `apk add` from the interactive shell at the bottom of the editor. If you use `apk add` from the interactive shell at the bottom of the editor, your score can be treated as zero on the private test cases conducted after submission, even if you were scoring points during the examination. Always perform `apk add` inside the makefile.

The library binaries will be stored in `/usr/lib` or `/lib`, and headers will be stored in `/usr/include`.

For example, to use libcurl, modify the makefile as follows, and write `#include <curl/curl.h>` inside `main.cpp`:

```
main: src/*.cpp makefile
    apk add curl-dev
    c++ -std=c++20 -o main src/*.cpp -lcurl
```

For example, to use one of the Boost libraries, `libboost_system`, modify the makefile as follows, and write `#include <boost/filesystem.hpp>` inside `main.cpp`:

```
main: src/*.cpp makefile
    apk add boost-dev
    c++ -std=c++20 -o main src/*.cpp -lboost_system
```

Copy

The available libraries are those provided as packages in Alpine Linux.

Submit Chal

main.cpp

```cpp
#include <iostream>
#include <string>
using namespace std;

int main(int argc, char *argv[]) {
  // このコードは標準入力と標準出力を用いたサンプルコードです。
  // このコードは好きなように編集・削除してもらって構いません。
  // ---
  // This is a sample code to use stdin and stdout.
  // Edit and remove this code as you like.

  string line;
  int index = 1;
  while (!cin.eof()) {
    getline(cin, line);
    cout << "line[" << index++ << "]:" << line << "\n";
  }
  return 0;
}
```

```
> echo "Hello track!"
Hello track!
>
> Enter command here.
```

Run & Save

F7 F8 F9 F10 F11

Accenture Japan - 2024 IT | Information Technology knowledge quiz 2-202...

Q1 /20

Choose the DNS server setting that will prevent an existing host and its IP address from being known.

A. ○ Prohibit third-party mail relay.

✓ B. ○ Control the accessing source of a zone transfer.

C. ○ Prohibit running a CGI program in other than a specified directory.

D. ○ Prohibit displaying file names existing in a directory.

Answer >

Q2 /20

A binary program failed to run because it crashed due to lack of memory. Which of the following is **NOT** an appropriate action to solve this issue?

A. ○ Use garbage collection to prevent certain kinds of memory leaks.

B. ○ Terminate other processes running at the same time.

✓ C. ○ Shorten the binary filename to save memory usage.

D. ○ Consider running the program on a device with more RAM capacity.

Answer >

Q3 /20

In UNIX-like OS, which is the correct meaning of the command `chmod u+x foo` ?

exeaut

user

A. ◯ Give execute permission to owner of the file named foo.

B. ◯ Give the group that the file named foo belongs to, execute permission of foo.

C. ◯ Give execute permission to all users who use the file foo.

D. ◯ Give execute permission to the owner of the file named foo and removes execute permission of all other users.

Answer >

Q4 /20

When browsing a website (`https://example.com`), the contents displayed on the smartphone and the computer are different. Which statement below is **not** the cause?

A. ◯ Because the content of the website is dynamically generated with JavaScript.

B. ◯ Because the content of the website is dynamically switched based on the user agent.

C. ◯ Because the content of the website was tampered by a communication path.

D. ◯ Because the content of the website was implemented with responsive design.

Answer >

Q5 /20

Which is the description of CSRF (Cross-Site Request Forgery)?

A. An attack that uses a vulnerability in a commercially available database management system to search for a database server to become a host, then repeatedly self-contamination to create a big surge in the website traffic.

B. An attack that embedded a malicious script on the website's visitor input form and executes on visitor's browser.

C. An attacks that embedded malicious input data to a web application form executing a query to tamper data or acquire user information illegally.

D. Attack that allows visitors to browse web pages with malicious scripts embedded in them, running unintended operations on other visitors browser.

Answer >

Q6 /20

Which is the cryptographic algorithm of forward secrecy (FS)?

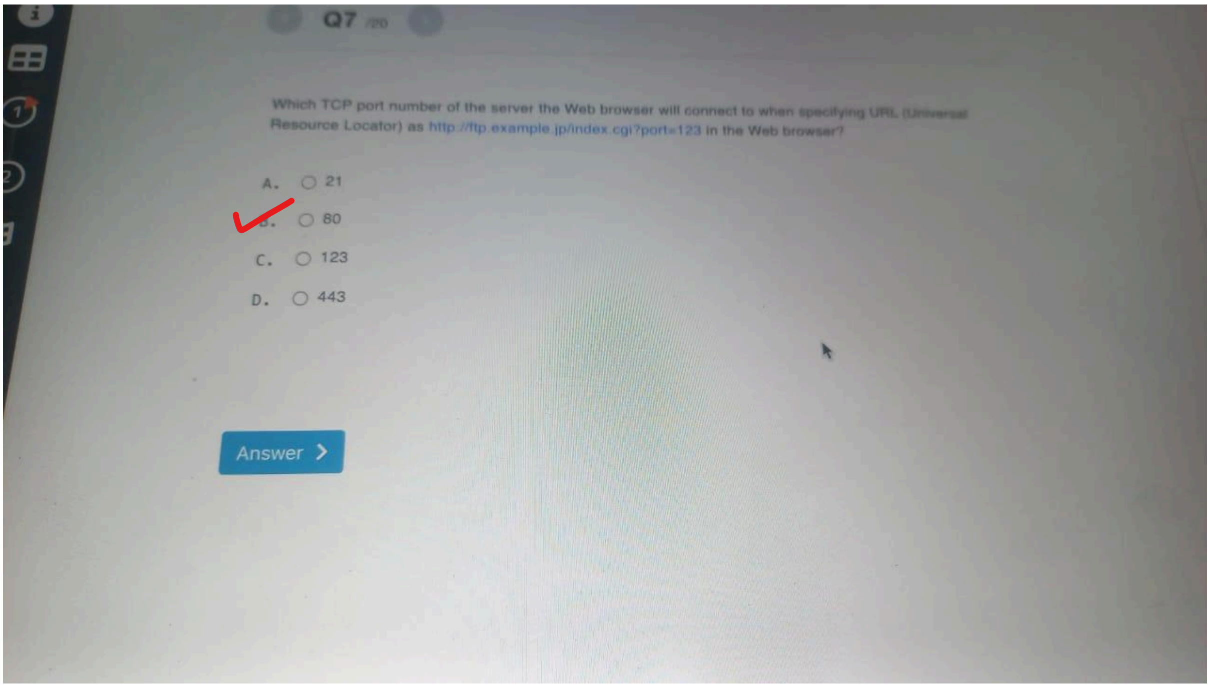
A. An improper way to manage keys will increase the risk of the key being compromised.

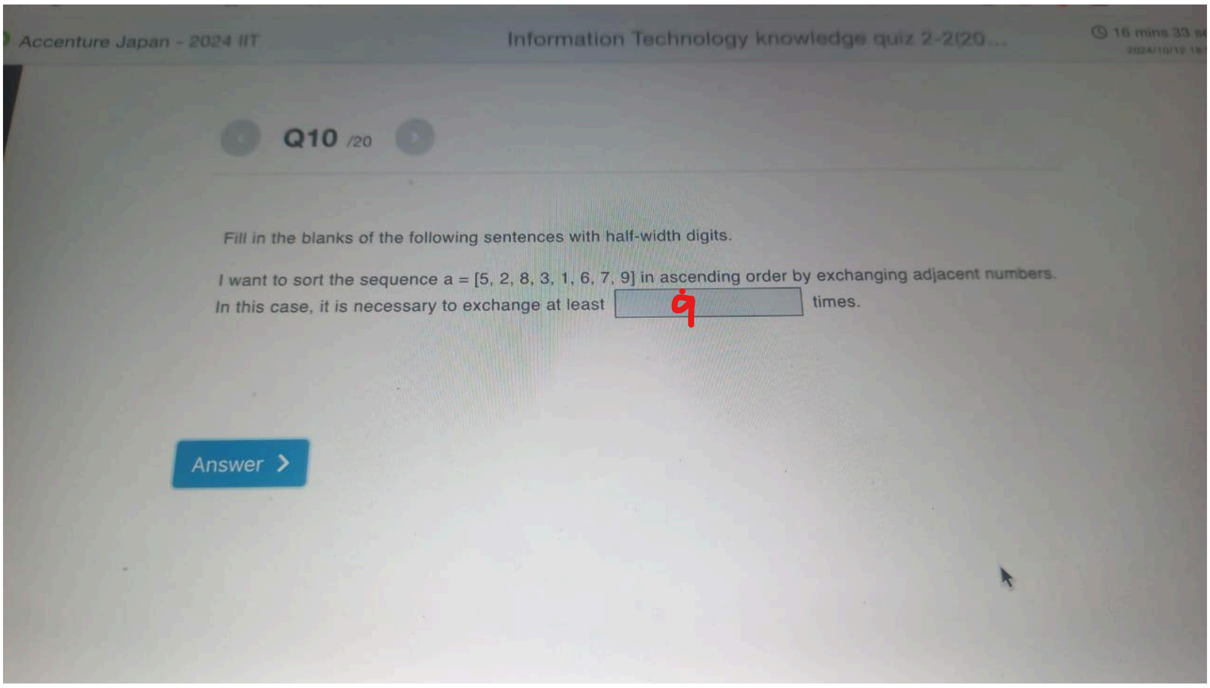
B. To ensure that even if the long-term encryption key is compromised in the future, the data encrypted with the short-term key will not be decrypted.

C. Using a computer with the high-performance specification to spend time and effort in increasing the reliability and strength of encryption.

D. Advances in technology will increase computing power and make it possible to decrypt in a realistic time.

Answer >


Q7 /20
Which TCP port number of the server the Web browser will connect to when specifying URL (Universal Resource Locator) as http://ftp.example.jp/index.cgi?port=123 in the Web browser?
A. 21
B. 80
C. 123
D. 443
Answer


Accenture Japan - 2024 IIT
Information Technology knowledge quiz 2-2(20...
16 mins 33 s
Q10 /20
Fill in the blanks of the following sentences with half-width digits.
I want to sort the sequence a = [5, 2, 8, 3, 1, 6, 7, 9] in ascending order by exchanging adjacent numbers.
In this case, it is necessary to exchange at least 9 times.
Answer

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Information Technology knowledge quiz 2-2(2024-10-9)

Q11 /20

There is a staircase of n steps and you are on the bottom of the staircase, on the 0-th step.
You can go up any of 1, 2, or 3 steps at a time.
Let us consider the problem of counting the number of step-taking patterns that will put you on exactly the n-th step from the bottom.

For instance, if there are $n = 4$ steps, there are 7 patterns as follows. Each number is the number of steps you take at one time.

- $1, 1, 1, 1.$
- $2, 1, 1.$
- $1, 2, 1.$
- $1, 1, 2.$
- $2, 2.$
- $1, 3.$
- $3, 1.$

For this problem, you implemented a function `solve` as follows.
Note that all the variables are multiple precision integers.

```
function rec(i):
  if i == 0:
    return 1
  else:
    ans = 0;
    if i >= 1:
      ans += rec(i-1)
    if i >= 2:
      ans += rec(i-2)
    if i >= 3:
      ans += rec(i-3)
    return ans

function solve(n):
  return rec(n)
```

However, the program is too slow and it does not return the answer for $n = 1000$ even if you wait a whole day. You want to speed up the program so that it returns the answer in one second even if n is in the range between 1000 and 10000.
Which one of the following correctly states the reason the function is too slow and the solution to fix the slowness?

A. ○ The three `if` statements in lines `6-11` cause the program to be slow. We can speed up the program by replacing them with a single for statement and calling `rec` inside the statement.

B. ○ When n is equal to 1000, an integer overflow occurs and causes an in[illegible] If we use a real data type instead of a multiple precision integer type, we can prev[illegible] infinite loop and then speed up the program.

C. ○ Although `rec(i)` always returns the same value, every time [illegible] the function three times recursively. That is why the program is so slow. We can [illegible] the program by storing the value `rec(i)` for each `i` when it is calculated [illegible] returning that value every time it is called thereafter.

D. ○ None of the above.

Answer ›

Q11 /20

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C. ○ Although `rec(i)` always returns the same value, every time `rec(i)` is called, it calls the function three times recursively. That is why the program is so slow. We can speed up the program by storing the value `rec(i)` for each `i` when it is calculated for the first time, and returning that value every time it is called thereafter.

D. ○ None of the above.

Answer >

Q12 /20

The following JavaScript function returns the number where the integer x occurs in the given array of integers a.

```
function find(a, x) {
    for (let i = 0; i < a.length; i++) {
        if (a[i] == x) {
            return i + 1;
        }
    }
    return -1;
}
```

Choose the correct statement regarding this function.

A. ○ If the length of array a is 0, the program terminates abnormally.

B. ○ If multiple occurrences of x are found in array a, all occurrences are returned.

C. ○ If x is not found in array a, the function returns -1.

D. ○ All of the above is incorrect.

Q13 /20

You are teaching beginners A, B, C at your company. The beginners wrote the following functions. These functions receive an array of integers `a` and outputs the sum of all elements in array `a`.

Choose all functions that work properly. At least 1 function works properly.

Beginner A

```
function sum(a) {
    let answer = 0;
    for (let i = 0; i < a.length; i++) {
        answer += a[i];
    }
    return answer;
}
```

Beginner B

```
function sum(a) {
    let answer = 0;
    for (let i = a.length; i > 0; i--) {
        answer += a[i];
    }
    return answer;
}
```

Beginner C

```
function sum3(a) {
    let answer = 0;
    let i = 0;
    while (1) {
        if (i == a.length) {
            break;
        }
        answer += a[i];
        i++;
    }
    return answer;
}
```

A. ☐ Beginner A's function

B. ☐ Beginner B's function

C. ☐ Beginner C's function

Answer ›

Q14 /20

This Java function receives an input of array of integers `arr` and outputs `true` if there are any neighboring array elements that have the same value in the array.

For example:
Given `arr = {0, 3, 2, 2, 6, -1, 3}`, the elements `2, 2` are next to each other and are the same value, so `true` is returned.
Given `arr = {0, 3, 2, 5, 6, -1, 3}`, there are no elements with the same values next to each other, so `false` is returned.

```
static boolean containConsecutive(int[] arr) {
    for (int i = 0; i < [ ① ]; i++) {
        if ([ ② ]) {
            return true;
        }
    }
    return false;
}
```

Choose the appropriate formula to enter into [①], [②] to make the function work.

A. ○ [①] `arr.length-1`, [②] `arr[i] == arr[i-1]`

B. ○ [①] `arr.length-1`, [②] `arr[i] == arr[i+1]`

C. ○ [①] `arr.length`, [②] `arr[i] == arr[i-1]`

D. ○ [①] `arr.length`, [②] `arr[i] == arr[i+1]`

E. ○ All the above are incorrect

2024/10/12 19:07

Q1 /20

Choose the DNS server setting that will prevent an existing host and its IP address from being known.

A. Prohibit third-party mail relay.

B. Control the accessing source of a zone transfer.

C. Prohibit running a CGI program in other than a specified directory.

D. Prohibit displaying file names existing in a directory.

Answer

Q3 /20

In UNIX-like OS, which is the correct meaning of the command `chmod u+x foo` ?

A. ○ Give execute permission to owner of the file named foo.

B. ○ Give the group that the file named foo belongs to, execute permission of foo.

C. ○ Give execute permission to all users who use the file foo.

D. ○ Give execute permission to the owner of the file named foo and removes execute permission of all other users.

Answer >

Q2 /20

A binary program failed to run because it crashed due to lack of memory. Which of the following is **NOT** an appropriate action to solve this issue?

A. ○ Use garbage collection to prevent certain kinds of memory leaks.

B. ○ Terminate other processes running at the same time.

C. ○ Shorten the binary filename to save memory usage.

D. ○ Consider running the program on a device with more RAM capacity.

Answer >

Q5 /20

Which is the description of CSRF (Cross-Site Request Forgery)?

A. An attack that uses a vulnerability in a commercially available database management system to search for a database server to become a host, then repeatedly self-contamination to crea te a big surge in the website traffic.

B. An attack that embedded a malicious script on the website's visitor input form and executes o n visitor's browser.

C. An attacks that embedded malicious input data to a web application form executing a query t o tamper data or acquire user information illegally.

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Answer

Q4 /20

When browsing a website (`https://example.com`), the contents displayed on the smartphone and the computer are different. Which statement below is **not** the cause?

A. ○ Because the content of the website is dynamically generated with JavaScript.

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D. ○ Because the content of the website was implemented with responsive design.

Q7 /20

Which TCP port number of the server the Web browser will connect to when specifying URL (Universal Resource Locator) as http://ftp.example.jp/index.cgi?port=123 in the Web browser?

A. ○ 21

B. ○ 80

C. ○ 123

D. ○ 443

Answer >

Q8 /20

Choose the correct description of database rollforward processing.

A. Restore using a journal file before updating against a physical failure.

B. Restore using a journal file after updating against a physical failure.

C. Restore using a journal file before updating against a logical failure.

D. Restore using a journal file after updating against a logical failure.

Answer

Q6 /20

Which is the cryptographic algorithm of forward secrecy (FS)?

A. ○ An improper way to manage keys will increase the risk of the key being compromised.

B. ○ To ensure that even if the long-term encryption key is compromised in the future, the data en crypted with the short-term key will not be decrypted.

C. ○ Using a computer with the high-performance specification to spend time and effort in increasi ng the reliability and strength of encryption.

D. ○ Advances in technology will increase computing power and make it possible to decrypt in a r ealistic time.

Answer >

Q9 /20

Mr. Aoki is planning to visit the shrine for New Years. However, the parking lot at the shrine is very small. Once you park your car, you cannot leave until all of the cars that came after you leave. Choose the data structure that best represents the situation in this parking lot.

A. ○ stack

B. ○ queue

C. ○ heap

D. ○ linked list

Answer

Q12 /20

The following JavaScript function returns the number where the integer `x` occurs in the given array of integers `a`.

```
function find(a, x) {
    for (let i = 0; i < a.length; i++) {
        if (a[i] == x) {
            return i + 1;
        }
    }
    return -1;
}
```

Choose the correct statement regarding this function.

A. ○ If the length of array `a` is 0, the program terminates abnormally.

B. ○ If multiple occurrences of `x` are found in array `a`, all occurrences are returned.

C. ○ If `x` is not found in array `a`, the function returns `-1`.

Answer >

Q1 N
Q2 N
Q3 No
Q4 No
Q5 Not
Q6 Not
Q7 Not A
Q8 Not A
Q9 Not A
Q10 Not A
Q11 Not A
Q12 Not An
Q13 Not An
Q14 Not Ans

Track Test

ks.run/abroad/exams/e3984d95d3dd43408f319854a9452cef/challenges/8327/2

2024 IIT

Information Technology knowledge quiz 2-1(2024-...

Q2 /20

In UNIX-like OS, the authority for a certain file varies depending on the user who uses it.
Which of the following description is correct?

A. ○ There is one owner in the file, and authority is given only to that user.

B. ○ There are multiple owners in the file, and privileges are granted only to those users.

C. ○ There is one owner and group in the file, and different authorities are given to the user who is the owner, the users belonging to the group and users of the other type.

D. ○ There are multiple owner and group in the file, and two types of users, owner and users belonging to the group.

Answer >

Q2 /20

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Answer >

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Q5 /20

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Information Technology knowledge qu

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